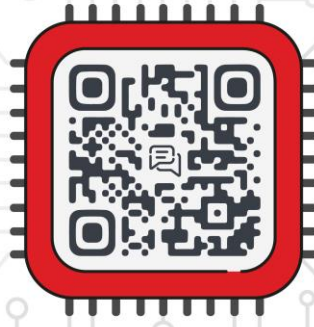




9



5



750 sp

نظري

كلية الهندسة المعلوماتية

السنة الثانية

Encoding



د. عماد الدين محمد

محتوى مجاني غير مخصص للبيع التجاري

RB Informatics; 22/6/2026

الاتصالات الرقمية

كما تبدأ الحكايات الجميلة، لا بد لها من ختام ... ولكن ختامنا اليوم هو فاتحة مستقبل مشرق لكم
جئنا ومعنا الشغف ... وها نحن نغادر ومعنا العلم والذكريات

أهلا بكم في ختام مشوارنا ♥

Content course:

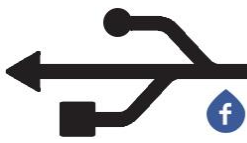
- Pulse-Code Modulation (PCM)
- PCM Stages: Sampling , quantization , quantization error and encoding
- Nyquist theorem
- Quantization
- PROS & CONS of PCM technology.
- Delta modulation
- Limitations & Errors



When the original data is analogue (such as a human voice) and we want to transmit it via a digital signal, we use a process called encoding. The basic and most well-known technology in this field is PCM (Pulse-Code Modulation).

Analog Data - Digital signal (Encoding):

Pulse-Code Modulation (PCM) is a digital method used to represent sampled analog signals, such as audio or voice. It serves as the standard for uncompressed digital audio in computers, CDs, and telephony. The conversion process translates continuous analog waves into a sequence of discrete binary numbers (0s and 1s).



The PCM process consists of four distinct steps:

1. Sampling:

- The continuous analog signal is measured at regular, uniform time intervals.
- According to the Nyquist theorem: the sampling frequency must be at least twice the maximum frequency of the analog signal to accurately capture and reconstruct it later.
- Experience has shown that if the frequency is less than double, parts of the information will be lost (such as parts of words falling out in the audio), while a higher frequency ensures the accuracy of the reconstruction.

2. Quantization:

- The exact voltage or amplitude of each sample is rounded off to the nearest predetermined discrete value within a specific range.

3. Quantization Error:

- The slight difference between the original analog value and the rounded digital value creates minor noise, which is minimized by increasing the number of available levels (e.g., going from 8-bit to 16-bit).

4. Encoding:

- Each discrete, quantized value is translated into a unique binary code (a sequence of bits). The resulting bit stream is then ready for transmission, storage, or digital processing.
- We calculate the number of bits needed based on the number of states using the law:

$$2^n \geq \text{Number of states or } n = \log_2(\text{Number of states})$$
- **Example:** if we have 100 states, we need 7bits because $2^7=128$ (enough) and $2^6 =64$ (not enough).

PCM'S PROS & CONS

pros	cons
Offers high signal fidelity (lossless quality) since it is uncompressed and provides excellent immunity to signal noise during transmission.	Requires significantly more bandwidth and storage space compared to compressed audio formats like MP3.

Applications

- **Audio:** Used in raw WAV and AIFF files, Compact Discs (CDs), and studio recording equipment.
- **Telecommunications:** Standard for digital voice transmission (like standard landline systems)

Delta modulation

Delta modulation is an analog-to-digital (ADC) and digital-to-analog (DAC) conversion technique that encodes a signal by transmitting only a single bit per sample. It tracks changes in the analog wave by stepping a stair-step approximation up or down, indicating whether the current sample is larger or smaller than the previous one.

How It Works

Delta modulation: works on a principle of difference rather than absolute value (making it a 1-bit version of Differential Pulse Code Modulation).

The Transmitter: The original analog signal is compared to an approximated staircase signal. If the analog signal is greater, it generates a positive difference (encoded as a digital 1). If it is smaller, it generates a negative difference (encoded as a digital 0).

The Receiver:

The receiver processes the stream of 1s and 0s using an accumulator, adding a step $+\Delta$ for every 1 and subtracting $-\Delta$ for every 0. A low-pass filter smooths this stepped approximation to reconstruct the original analog signal.

Advantages

Simplicity: The transmitter and receiver require very basic hardware (a comparator, a quantizer, and a delay circuit), making implementation cheaper and less power-consuming than complex methods like Pulse Code Modulation (PCM).

Low Bandwidth: Because it encodes the signal using only 1 bit per sample, it uses a significantly lower transmission bandwidth than multi-bit encoding schemes.

Summary:

Advantages: Simple hardware, and it sends fewer bits compared to PCM, which may need 7 or 8 bits per sample.

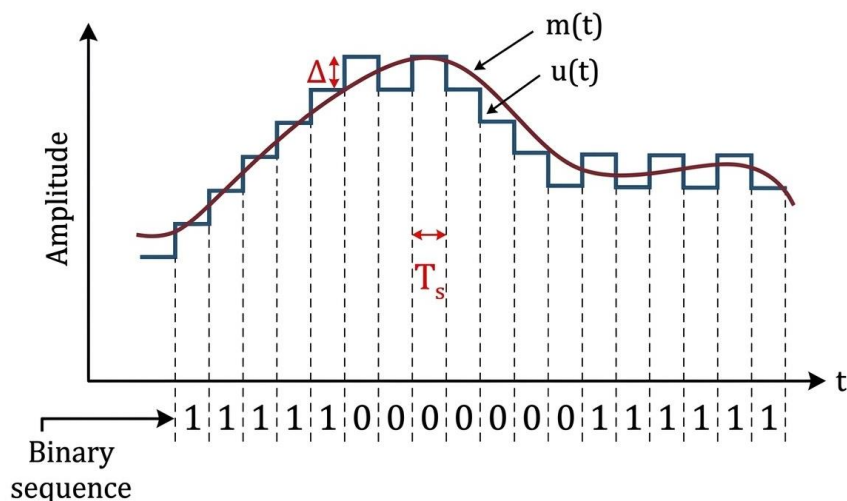
Limitations & Errors

Because delta modulation relies on a fixed step size (Δ) between samples, it is prone to two primary types of errors:

Slope Overload Distortion: This occurs when the analog signal changes too rapidly for the staircase approximation to keep up. The step size is too small to follow the fast rising or falling edges of the signal.

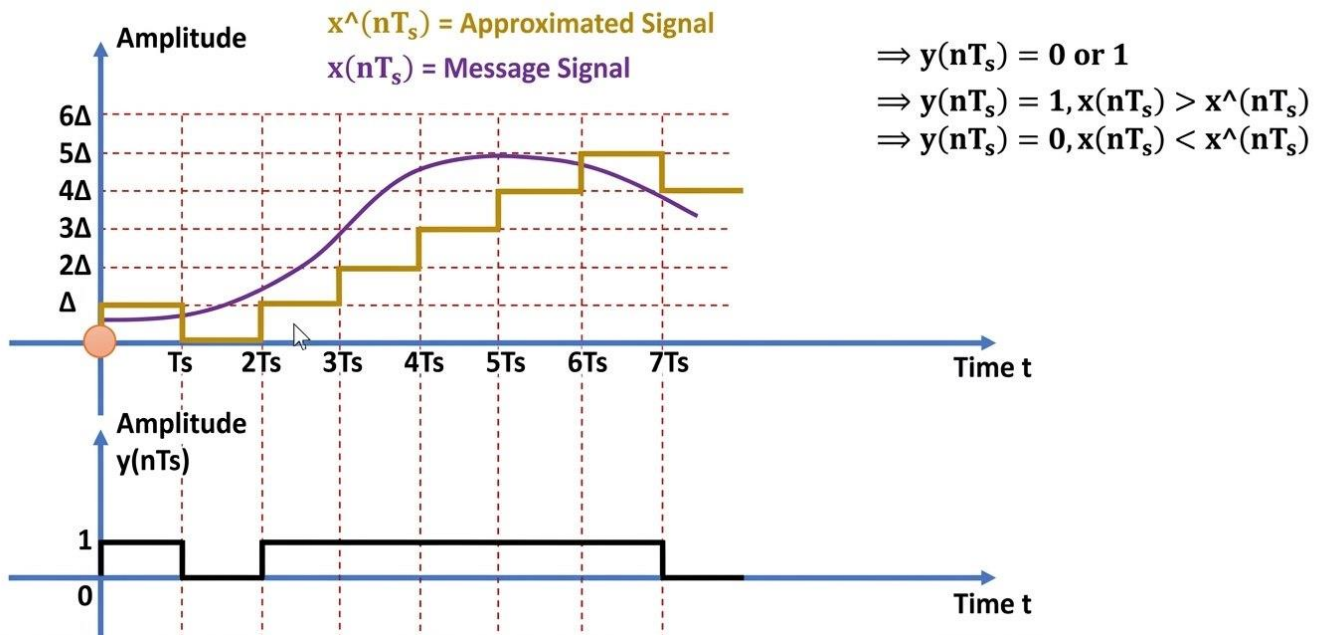
Granular (Idle) Noise: This occurs when the original signal is changing very slowly or is flat. In this case, the staircase approximation overshoots and oscillates up and down around the actual signal, creating background noise.

To overcome these limitations, an enhanced version called **Adaptive Delta Modulation (ADM)** is often used. ADM automatically varies the step size—making it larger during rapid signal changes to prevent slope overload, and smaller during slow changes to reduce granular noise.



Waveform representation of Delta modulation

Waveform of Delta Modulation



If Message Signal $x(nT_s)$ is less than Approximated Signal $\hat{x}(nT_s)$, then reduce the size of the Approximated signal by step size Δ for the next T_s duration, and vice versa



The end ...